Scalable fit of the output-error sensitivity of a Clements unitary mesh vs. effective insertion-loss spread (fixed $N = 100$, varying μ_L)

FIT on the small range $\sigma_{\text{eff}} \in [0, 0.263]$ dB (panels A and B together)

Model (overlap-compatible, scalable in N/N_0): $E(\mu_L, \sigma_{\text{eff}}, N) = \sqrt{2}\,\sqrt{1 - e^{-(u_\mu^2 + u_\sigma^2)}}$, $u_\sigma = \dfrac{s_\sigma(N)\,\sigma_{\text{eff}}}{\sqrt{2}}$, $s_\sigma(N) = s_{\sigma,0}\,(N/N_0)^{p_\sigma}$, $N_0 = 100$

u_μ = mean-loss term of the loss fit (μ_L only); σ_{eff} = swept variable = realised std of $L \sim \text{TruncNorm}(\mu_L, \sigma_L^2; 0, 2\mu_L)$ dB (nominal σ_L derived in the sweep; no conversion and no fitted parameter here)

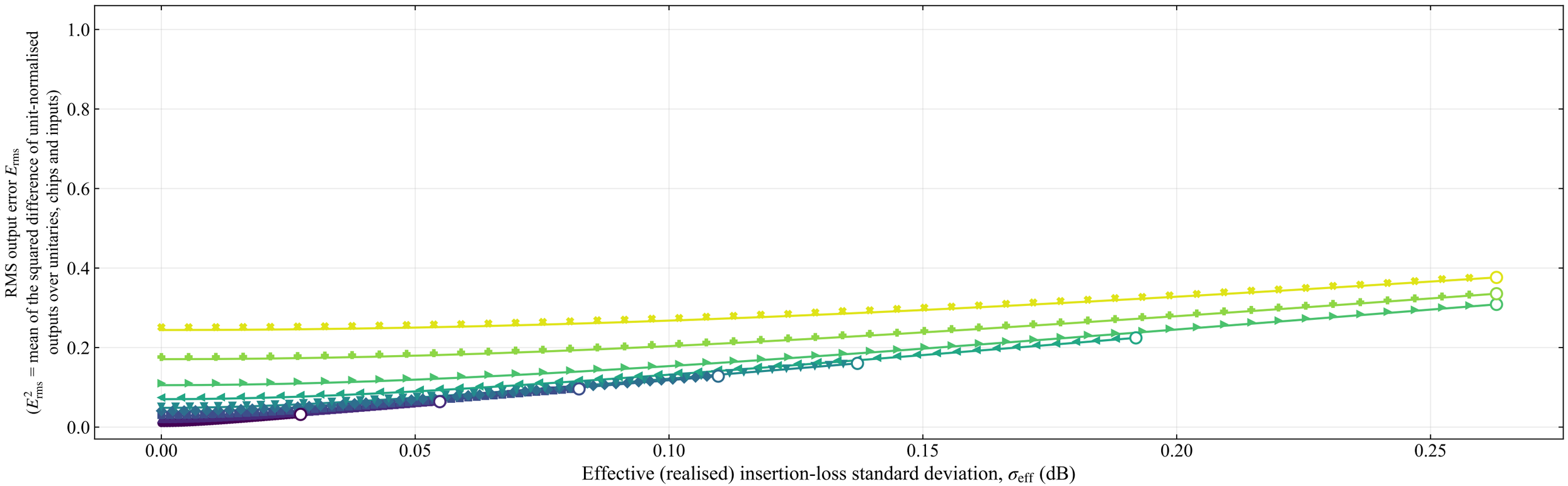
$s_{\sigma,0}$ = spread sensitivity at $N_0 = 100$ (dB^{-1}) | p_σ = network-size exponent of s_σ

Best-fit values (small range, panels A+B, equal weight per point; ± = bootstrap s.d. over unitaries): $s_{\sigma,0} = 1.11787 \pm 0.00025$ dB^{-1}, $p_\sigma = 0.51651 \pm 0.00017$

Quality on this panel: full surrogate (lines): RMSE = 0.00314, MAE = 0.00272, $R^2 = 0.99898$, max|ΔE| = 0.0070; off-plateau ($E \le 1.2$, 450 pts): RMSE = 0.00314

spread term alone, anchored at the measured $E(\sigma_{\text{eff}} = 0)$: RMSE = 0.00054, $R^2 = 0.99997$, max|ΔE| = 0.0018 | mean-loss term only (no spread term): RMSE = 0.06506, $R^2 = 0.56162$

Haar-random unitary U (single Clements mesh) | 7 input types × 10 draws, pooled | one forward pass $y = Ux$ | $E = \|\hat{y}^{\text{hw}} - \hat{y}^{\text{ideal}}\|_2$ | 25 unitaries × 30 chips | 50 points per curve



Mean loss μ_L ($N = 100$) | each curve $\sigma_{\text{eff}} \in [0, \sigma_{\text{top}}(\mu_L)]$, $\sigma_{\text{top}} = \min(0.95\,\mu_L/\sqrt{3},\ 0.263$ dB) | markers: data, lines: full surrogate, circles: end of curve

- $\mu_L = 0.05$ dB, $\sigma_{\text{eff}} \in [0, 0.02742]$ dB
- $\mu_L = 0.1$ dB, $\sigma_{\text{eff}} \in [0, 0.05485]$ dB
- $\mu_L = 0.15$ dB, $\sigma_{\text{eff}} \in [0, 0.08227]$ dB
- $\mu_L = 0.2$ dB, $\sigma_{\text{eff}} \in [0, 0.1097]$ dB
- $\mu_L = 0.25$ dB, $\sigma_{\text{eff}} \in [0, 0.1371]$ dB
- $\mu_L = 0.35$ dB, $\sigma_{\text{eff}} \in [0, 0.192]$ dB
- $\mu_L = 0.5$ dB, $\sigma_{\text{eff}} \in [0, 0.263]$ dB
- $\mu_L = 0.75$ dB, $\sigma_{\text{eff}} \in [0, 0.263]$ dB
- $\mu_L = 1$ dB, $\sigma_{\text{eff}} \in [0, 0.263]$ dB